MATHEMATICS

SEQUENCE

&

PROGRESSION

THEORY SHEET

ENGLISH

MATRIX

Piprali Road, Sikar
Ph. 01572-241911, www.matrixedu.in

1. INTRODUCTION OF SEQUENCE

A succession of terms a_1, a_2, a_3, a_4..... formed according to some rule or law.

Examples are :1, 4, 9, 16, 25.....; –1, 1, –1, 1,.....; $\frac{x}{1!}, \frac{x^2}{2!}, \frac{x^3}{3!}, \frac{x^4}{4!}$, ...

General Term

It is an expression from which we can find all the terms of the sequence.

Example : $T_n = n^2 + 2n$

Sum of first n terms

$S_n = T_1 + T_2 + ...+ T_{n-1} + T_n$

$S_{n-1} = T_1 + T_2 + ...+ T_{n-1}$

$T_n = S_n - S_{n-1}$

Q.1 $S_n = 2n^2 + 3n$ find T_n

Ans $T_n = 4n + 1$

Sol. $T_n = S_n - S_{n-1}$

$= (2n^2 + 3n) - [2(n-1)^2 + 3(n-1)]$

$T_n = 4n + 1$

Important Sequences

(1) Arithmetic Progression (AP)

(2) Geometric Progression (GP)

(3) Harmonic Progression (HP)

(4) Arithmetic Geometric Progression (AGP)

(5) Miscellaneous Sequences

2.1 ARITHMETIC PROGRESSION

A.P. is a sequence whose successive terms are obtained by adding a fixed number 'd' to the preceding terms. This fixed number 'd' is called the common difference.

Standard Form:

a, a + d, a + 2 d,..... ,a + (n – 1) d,.....

where a = first term & d = common difference

e.g. – 4, – 1, 2, 5 .....

n^{th} term of an A.P. :

Let 'a' be the first term and 'd' be the common difference of an A.P., then

$T_n = a + (n - 1)\, d$, where $d = T_n - T_{n-1}$

Q.1 For following arithmetic progression, find T_n

(i) 2, 5, 8, ... (ii) 10, 6, 2, ...

Ans. (i) $T_n = 3n - 1$ (ii) $T_n = 14 - 4n$

Sol. (i) $T_n = 2 + (n - 1)3$

$= 3n - 1$

(ii) $T_n = 10 + (n - 1)(-4)$

$= 10 - 4n + 4$

$= 14 - 4n.$

Q.2 If t_{54} of an A.P. is -61 and $t_4 = 64$, find t_{10}.

Ans 49

Sol. Let a be the first term and d be the common difference

so $t_{54} = a + 53d = -61$ .....(i)

and $t_4 = a + 3d = 64$ .....(ii)

equation (i) – (ii) we get

$\Rightarrow 50d = -125$

$d = -\frac{5}{2}$

$\Rightarrow a = \frac{143}{2}$ so $t_{10} = \frac{143}{2} + 9\left(-\frac{5}{2}\right) = 49$

Q.3 For an arithmetic progression, if $T_m = n$, $T_n = m$, $m \neq n$ find T_p.

Ans. $T_p = m + n - p$

Sol.

$$a + (m-1)d = n$$
$$a + (n-1)d = m$$
$$(m-n)d = n - m$$

$d = -1$

$a = n - (m - 1)(-1)$

$a = n + m - 1$

$T_p = a + (p - 1)d$

$= n + m - 1 - p + 1$

$= m + n - p.$

Q.4 For an arithmetic progression, if $T_p = \frac{1}{q}$, $T_q = \frac{1}{p}$, find T_{pq} $(p \neq q)$.

Ans. $T_{pq} = 1$

Sol. $a + (p-1)d = \frac{1}{q}$

$$a+(q-1)d=-\frac{1}{p}$$

– –

$$(p-q)d=\frac{p-q}{pq}$$

$$d=\frac{1}{pq}$$

$$a=\frac{1}{q}-(p-1)\frac{1}{pq}$$

$$a=\frac{1}{q}-\frac{1}{q}+\frac{1}{pq}$$

$$a=\frac{1}{pq}$$

$T_{pq} = a + (pq - 1)d$

$$=\frac{1}{pq}+(pq-1)\frac{1}{pq}$$

$$=\frac{1}{pq}+1-\frac{1}{pq}\Rightarrow 1$$

Q.5 Find the number of terms in the AP 4, 12, 20, ...,108.

Ans 14

Sol. $a = 4, d = 8$ so $108 = 4 + (n - 1)8$

$\Rightarrow n = 14$

2.2 SUM OF FIRST 'n' TERMS OF AN AP

If a is first term and d is common difference

$S_n = a + (a + d) + (a + 2d) + ... + a+(n–1)d$

$S_n = (a + (n –1)d)+ (a + (n –2)d) + (a + (n–3)d) + ... + a$

add $2S_n = 2a + (n –1)d + 2a + (n –1)d + 2a + (n–1)d + ... + 2a + (n–1)d$

$2S_n = n (2a + (n–1)d)$

$$S_n = \frac{n}{2}[2a + (n - 1) d]$$

$$= \frac{n}{2}[a + \ell\ ]$$ (Where ℓ is the last term)

2.3 HIGHLIGHTS OF AN AP

1. The common difference can be zero, positive or negative.
   (i) $d > 0 \Rightarrow$ Increasing AP
   (ii) $d = 0 \Rightarrow$ All terms are equal

(iii) $d < 0 \Rightarrow$ Decreasing AP

2. If a, b, c are in AP, then $2b = a + c$ and converse is also true

3. Three consecutive numbers in A.P. : $a - d, a, a + d$

   Four consecutive numbers in A.P. : $a - 3d, a - d, a + d, a + 3d$

   Five consecutive numbers in A.P. : $a - 2d, a - d, a, a + d, a + 2d$

4. The sum of the two terms of an AP equidistant from the beginning & end is constant and equal to the sum of first & last terms. $\Rightarrow T_k + T_{n-k+1} =$ constant $= a + \ell$.

5. n^{th} term of an A.P. is of the form An + B i.e. a linear expression in 'n', in such a case the coefficient of n is the common difference of the A.P. i.e. A.

6. Sum of first 'n' terms of an A.P. is of the form $An^2 + Bn$ i.e. a quadratic expression in 'n', in such case the common difference is twice the coefficient of n^2. i.e. 2A

7. If each term of an A.P. is increased, decreased, multiplied or divided by the same nonzero number, then the resulting sequence is also an A.P.

8. If corresponding terms of two APs or being added or subtracted then the resulting sequence will also be an AP.

Q.1 First & last terms of an AP are 5 and 45 respectively & its sum is 400. Find common difference.

Ans. $\frac{8}{3}$

Sol. $a = 5 \qquad T_n = 45 \qquad S_n = 400$

$$S_n = \frac{n}{2}[a + \ell]$$

$$400 = \frac{n}{2}[5 + 45]$$

$$\frac{800}{50} = n$$

$16 = n.$

$T_n = a + (n - 1)d$

$45 = 5 + 15d$

$$\frac{40}{15} = d$$

$$\frac{8}{3} = d$$

Q.2 For an arithmetic progression, if $T_1 + T_2 + T_3 = 15$, $T_1 . T_2 . T_3 = 105$ find T_n.

Ans $T_n = 2n + 1$ or $T_n = 9 - 2n$

Sol. $T_1 = a - d$

$T_2 = a$

$T_3 = a + d$

$3a = 15$

$a = 5$

$(5 - d)(5)(5 + d) = 105$

$25 - d^2 = 21$

$4 = d^2$

$\pm 2 = d$

$d = 2$	$d = -2$
$T_1 = 5 - 2 = 3$	$T_1 = 5 - (-2) = 7$
$T_2 = 5$	$T_2 = 5$
$T_3 = 5 + 2 = 7$	$T_3 = 5 - 2 = 3$
3, 5, 7	7, 5, 3 .....
$T_n = 3 + (n - 1)(2)$	$T_n = 7 + (n - 1)(-2)$
$= 2n + 1$	$= 9 - 2n$

Q.3 For an arithmetic progression: $20,\ 19\frac{1}{3}\ ,\ 18\frac{2}{3}\ ,$ .....

(i) Find number of positive terms. (ii) Find smallest positive term. (iii) If $S_n = 300$ find n.

Ans (i) 30 (ii) $\frac{2}{3}$ (iii) n = 25 or 36

Sol. $a = 20 \quad d = -\frac{2}{3}$

Let $T_n = 0$

$a + (n - 1)d = 0$

$$20 + (n-1)\left(\frac{-2}{3}\right) = 0$$

$$(n-1) = \frac{-20 \times 3}{-2}$$

$n - 1 = 30$

$n = 31$

(i) No. of the terms = 30

(ii) As $T_n = 0$ is possible in series.

$\therefore\ T_{n-1}$ term will be $\frac{2}{3}$

$\therefore$ Smallest +ve term $= \frac{2}{3}$

(iii) $S_n = 300$ $\quad n = ?$

$a = 20 \quad d = \frac{-2}{3}$

$$300 = \frac{n}{2}\left[2(20) + (n-1)\left(-\frac{2}{3}\right)\right]$$

$$300 = n\left[20 - \frac{n+1}{3}\right]$$

$$300 = n\left[\frac{60 - n + 1}{3}\right]$$

$$300 = \frac{n}{3}[61 - n]$$

$900 = 61n - n^2$

$n^2 - 61n + 900 = 0$

$(n - 36)(n - 25) = 0$

$n = 36$ or $\quad n = 25$

Q.4 For an arithmetic progression: 2, 5, 8,.....find $\sum_{r=11}^{20} T_r$.

Ans 455

Sol. $a = 2 \quad d = 3$

$T_{11} + T_{12} + T_{13}.....T_{20} \Rightarrow S_{20} - S_{10}$

$$\frac{20}{2}[2(2) + 19(3)] - \frac{10}{2}[2(2) + 9(3)]$$

$= 10[4 + 57] - 5[4 + 27]$

$= 610 - 155$

$= 455$

Q.5 For an arithmetic progression: $T_m = \frac{1}{n}$, $T_n = \frac{1}{m}$ & $m \neq n$ prove that $S_{mn} = \frac{mn+1}{2}$.

Ans. $= \frac{mn+1}{2}$

Sol. $a + (m-1)d = \frac{1}{n}$

$$a+(n-1)d=\frac{1}{m}$$

$$-\quad -\qquad -$$

$$(m-n)d=\frac{m-n}{mn}$$

$$d=\frac{1}{mn}$$

$$a=\frac{1}{n}-(m-1)\frac{1}{mn}$$

$$a=\frac{1}{n}-\frac{1}{n}+\frac{1}{mn}$$

$$a=\frac{1}{mn}$$

$$S_{mn}=\frac{mn}{2}\left[2\left(\frac{1}{mn}\right)+(mn-1)\left(\frac{1}{mn}\right)\right]$$

$$=\frac{mn}{2}\left[\frac{2}{mn}+1-\frac{1}{mn}\right]$$

$$=\frac{mn}{2}\left[\frac{mn+1}{mn}\right]$$

$$=\frac{mn+1}{2}$$

Q.6 For an arithmetic progression if $S_p = q$, $S_q = p$; $p \neq q$ prove that $S_{p+q} = -(p + q)$.

Ans. $-(p + q)$

Sol. $\frac{p}{2}[2a+(p-1)d]=q$ $\qquad$ $\frac{q}{2}[2a+(q-1)d]=p$

$p[2a + (p - 1)d] = 2q$

$q[2a + (q - 1)d] = 2p$

$-\qquad\qquad -$

$2ap - 2aq + p^2d - pd - q^2d + qd = 2(q - p)$

$(q - p)(2a) + (q + p)(p - q)d - (p - q)d = -2(p - q)$

$2a + (p + q - 1)d = -2$

$$S_{p+q}=\frac{p+q}{2}\left[2a+(p+q-1)d\right]$$

$= \frac{p+q}{2}(-2) \Rightarrow -(p+q)$

Q.7 Find sum of all integers from 1 to 100 divisible by 2 or 3.

Ans 3417

Sol.

Sum of 2	Sum of 3
2, 4 100	3, 6, 9 99
$n-1=\frac{100-2}{2}$	$n-1=\frac{99-3}{3}$
n = 50	n = 33
$S_{50}=\frac{50}{2}[2(2)+49(2)]$	$S_{33}=\frac{33}{2}[2(3)+32(3)]$
= 50 × 51	= 33(51)
= 2550	= 1683

6, 12, 18.....96

$n-1 = \frac{96-6}{6}$

n = 16

$S_{16}=\frac{16}{2}[2(6)+15(6)]$

= 16[6 + 45]
= 16 × 51
= 816
$S_2 + S_3 - S_6 = 2550 + 1683 - 816$
= 4233 – 816
= 3417

Q.8 If $\log_3 2, \log_3(2^x-5), \log_3\left(2^x-\frac{7}{2}\right)$ are in AP then find x. IIT'90

Ans x = 3

Sol $\log_3(2^x-5) - \log_3 2 = \log_3\left(2^x-\frac{7}{2}\right) - \log_3(2^x-5)$; $\log_3\left(\frac{2^x-5}{2}\right) = \log_3\left(\frac{2^x-\frac{7}{2}}{2^x-5}\right)$

$\frac{2^x-5}{2} = \frac{2^x-\frac{7}{2}}{2^x-5}$; $\left(2^x-5\right)^2 = 2^{x+1}-7$

$2^{2x} - 10.2^x + 25 = 2.2^x - 7$; put $2^x = y$

$$y^2 - 12y + 32 = 0 \Rightarrow y = 8 \text{ or } 4\text{; hence } x = 3 \text{ (as } x = 2 \text{ is rejected)}$$

Q.9 If $S_n = 2n^2 + 4n$. Find T_n.

Ans. $T_n = 4n + 2$

Sol. $T_n = S_n - S_{n-1}$

$T_n = 2n^2 + 4n - [2(n-1)^2 + 4(n-1)]$

$= 2n^2 + 4n - [2(n^2 - 2n + 1) + 4n - 4]$

$= 2n^2 + 4n - 2n^2 + 4n - 2 - 4n + 4$

$= 4n + 2$

Q.10 The sum of n terms of two A.Ps. are in ratio $\frac{7n+1}{4n+27}$. Find the ratio of their 11th terms.

Ans $\frac{4}{3}$

Sol. Let a_1 and a_2 be the first terms and d_1 and d_2 be the common differences of two A.P.s respectively, then

$$\frac{\frac{n}{2}[2a_1 + (n-1)d_1]}{\frac{n}{2}[2a_2 + (n-1)d_2]} = \frac{7n+1}{4n+27}$$

$$\Rightarrow \frac{a_1 + \left(\frac{n-1}{2}\right)d_1}{a_2 + \left(\frac{n-1}{2}\right)d_2} = \frac{7n+1}{4n+27}$$

For ratio of 11th terms

$\frac{n-1}{2} = 10 \qquad \Rightarrow n = 21$

so ratio of 11th terms is $= \frac{7(21)+1}{4(21)+27} = \frac{148}{111} = \frac{4}{3}$

Q.11 Find value of $100^2 - 99^2 + 98^2 - 97^2 + \ldots\ldots\ldots\ldots + 2^2 - 1^2$

Ans 5050

Sol. $(100 - 99)(100 + 99) + (98 - 97)(98 + 97) + \ldots\ldots\ldots + (2 + 1)(2 - 1)$

$199 + 195 + 191 \ldots\ldots\ldots\ldots + 3$

$a = 199 \qquad d = -4 \qquad T_n = 3$

$\frac{3 - 199}{-4} = n - 1$

$49 + 1 = n$

$50 = n$

$$S_{50} = \frac{50}{2}[199+3]$$

$= 25 \times 202$

$= 50 \times 101$

$= 5050$

Q.12 Roots of equation $x^3 - 12x^2 + 39x - k = 0$ are in AP then find k and roots.

Ans. $k = 28$; roots are $\{1, 4, 7\}$

Sol. $x^3 - 12x^2 + 39x - k = 0$ $\begin{matrix}\alpha \\ \beta \\ \gamma\end{matrix}\}$ AP

$2\beta = \alpha + \gamma.$

$8 = \alpha + \gamma$

$\alpha + \beta + \gamma = 12$ $\quad$ $\alpha\beta + \beta\gamma + \gamma\alpha = 39$ $\quad$ $\alpha\beta\gamma = k$

$3\beta = 12$ $\quad$ $4(\alpha + \gamma) + \alpha\gamma = 39$ $\quad$ $4 \times 7 = k$

$32 + \alpha\gamma = 39$ $\quad$ $28 = k$

$\beta = 4$ $\quad$ $\alpha\gamma = 7$

$x^2 - 8x + 7 = 0$ $\begin{matrix}\alpha \\ \gamma\end{matrix}$

$(x - 7)(x - 1) = 0$

$x = 7$ or $x = 1$

Q.13 If sum of roots of quadratic equation $ax^2 + bx + c = 0$ is equal to sum of squares of their reciprocals, prove that bc^2, ca^2, ab^2 are in AP.

Ans.

Sol. $ax^2 + bx + c = 0$ $\begin{matrix}\alpha \\ \beta\end{matrix}$

$$\alpha + \beta = -\frac{b}{a} \quad \alpha\beta = \frac{c}{a} \qquad \alpha^2 + \beta^2 = \frac{b^2}{a^2} - \frac{2c}{a} \Rightarrow \frac{b^2 - 2ac}{a^2}$$

$$\alpha + \beta = \frac{1}{\alpha^2} + \frac{1}{\beta^2}$$

$$\alpha + \beta = \frac{\alpha^2 + \beta^2}{(\alpha\beta)^2}$$

$$\frac{-b}{a} = \frac{\frac{b^2 - 2ac}{a^2}}{\frac{c^2}{a^2}}$$

$$\frac{-b}{a} = \frac{b^2 - 2ac}{c^2}$$

$-bc^2 = ab^2 - 2a^2c$

$2a^2c = ab^2 + bc^2$

$ab^2, a^2c, bc^2 \rightarrow$ AP

Q.14. AP_1: 1, 5, 9, ...2001;

AP_2: 1, 7, 13, ...2101 find number of common terms.

Ans. 167

Sol. $a_1 = 1$ $d_1 = 4$ $a_2 = 1$ $d_2 = 6$

Common term d (LCM d_1 d_2) = 12

1, 13, ...

a = 1 d = 12 Let $T_n = 2001$

$T_n = a + (n - 1)d$

$2001 = 1 + (n - 1)\ 12$

$$\left[\frac{200.0}{12} + 1\right] = n$$

$[166.6 + 1] = n$

$[167.6] = n$

$n = 167$

Q.15 If a, b, & c are in A.P. then.

(i) Prove that b + c, c + a, a + b are in AP.

(ii) Prove that $(b + c)^2 - a^2$, $(c + a)^2 - b^2$, $(c + a)^2 - c^2$ are in AP.

Ans.

Sol. (i) b + c, c + a, a + b

a + b + b + c 2(a + c)

a + 2b + c

a + a + c + c

2a + 2c

2(a + c) = 2(a + c)

(ii) $(b + c)^2 - a^2$, $(c + a)^2 - b^2$, $(a + b)^2 - c^2$

(b + c + a) (b + c – a), (c + a + b) (c + a – b), (a + b + c) (a + b – c)

Divide by (a + b + c)

b + c – a, c + a – b, a + b – c

Subtract (a + b + c)

– 2a, –2b, –2c

Divide by –2

a, b, c $\rightarrow$ A.P.

Q.16 If a^2, b^2, c^2 are in AP.

(i) Prove that $\frac{1}{b+c}, \frac{1}{c+a}, \frac{1}{a+b}$ are in AP.

(ii) Prove that $\frac{a}{b+c}, \frac{b}{c+a}, \frac{c}{a+b}$ are in AP.

Sol a^2, b^2, c^2 are in A.P. (given)

(i) $a^2 + ab + bc + ca$, $b^2 + ab + bc + ca$, $c^2 + ab + bc + ca$ are in A.P.

$(a + b)(a + c), (b + c)(b + a), (c + a)(c + b)$ are in A.P.

or $\frac{1}{b+c}, \frac{1}{c+a}, \frac{1}{a+b}$ are in A.P.

(ii) TPT $\frac{a}{b+c}, \frac{b}{c+a}, \frac{c}{a+b}$ are in A.P.

or TPT $\frac{a}{b+c} + 1, \frac{b}{c+a} + 1, \frac{c}{a+b} + 1$ are in A.P.

or TPT $\frac{a+b+c}{b+c}, \frac{a+b+c}{c+a}, \frac{a+b+c}{a+b}$ are in A.P.

or TPT $\frac{1}{b+c}, \frac{1}{c+a}, \frac{1}{a+b}$ are in A.P.

Q.17 If $a_1, a_2, \ldots\ldots a_n$ are in AP then.

(i) Prove that $\frac{1}{a_1a_2} + \frac{1}{a_2a_3} + \frac{1}{a_3a_4} + \ldots\ldots\ldots\ldots + \frac{1}{a_{n-1}a_n} = \frac{n-1}{a_1a_n}$.

(ii) $\frac{1}{a_1a_n} + \frac{1}{a_2a_{n-1}} + \frac{1}{a_3a_{n-2}} + \ldots\ldots\ldots \frac{1}{a_na_1} = \frac{2}{a_1+a_n}\left(\frac{1}{a_1} + \frac{1}{a_2} + \ldots\ldots\ldots \frac{1}{a_n}\right)$.

Ans.

Sol. (i) $\frac{1}{a_1a_2} + \frac{1}{a_2a_3} + \frac{1}{a_3a_4} + \cdots + \frac{1}{a_{n-1}a_n} = \frac{n-1}{a_1a_n}$

$$L.H.S = \frac{1}{a}\left[\frac{d}{a_1a_2}+\frac{d}{a_2a_3}+\frac{d}{a_3a_4}+\cdots\frac{d}{a_{n-1}a_n}\right]$$

$$=\frac{1}{d}\left[\frac{a_2-a_1}{a_1a_2}+\frac{a_3-a_2}{a_2a_3}+\frac{a_4-a_3}{a_3a_4}\cdots\frac{a_n-a_{n-1}}{a_{n-1}a_n}\right]$$

$$=\frac{1}{d}\left[\frac{1}{a_1}-\frac{1}{a_2}+\frac{1}{a_2}-\frac{1}{a_3}+\frac{1}{a_3}-\frac{1}{a_4}\cdots+\frac{1}{a_{n-1}}-\frac{1}{a_n}\right]$$

$$=\frac{1}{d}\left[\frac{a_n-a_1}{a_1a_n}\right]$$

$$=\frac{1}{d}\left[\frac{a+(n-1)d-a}{a_1a_n}\right]\Rightarrow\frac{n-1}{a_1a_n}$$

(ii) $$\frac{1}{a_1a_n}+\frac{1}{a_2a_{n-1}}+\frac{1}{a_3a_{n-2}}+\cdots\frac{1}{a_na_1}=\frac{2}{a_1+a_n}\left[\frac{1}{a_1}+\frac{1}{a_2}+\cdots\frac{1}{a_n}\right]$$

$$L.H.S=\frac{1}{a_1+a_n}\left[\frac{a_1+a_n}{a_1a_n}+\frac{a_2+a_{n-1}}{a_2a_{n-1}}+\cdots\frac{a_n+a_1}{a_1a_n}\right]$$

$$=\frac{1}{a_1+a_n}\left[\frac{1}{a_1}+\frac{1}{a_n}+\frac{1}{a_2}+\frac{1}{a_{n-1}}+\cdots\frac{1}{a_1}+\frac{1}{a_n}\right]$$

$$=\frac{2}{a_1+a_n}\left[\frac{1}{a_1}+\frac{1}{a_2}+\frac{1}{a_3}\cdots\frac{1}{a_n}\right]$$

Q.18 Prove that $\sqrt{2},\sqrt{3},\sqrt{5}$ can not be any three terms of AP.

Sol Let $\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$ are p^{th} , q^{th} and r^{th} terms.

Hence $\sqrt{2} = a + (p-1)d$; $\sqrt{3} = a + (q-1)d$; $\sqrt{5} = a + (r-1)d$

$\therefore$ $\dfrac{\sqrt{5}-\sqrt{3}}{\sqrt{3}-2}=\dfrac{r-q}{q-p}$; RHS is rational and LHS is irrational

Q.19 Let S_1 be the sum of first 2n terms of an arithmetic progression. Let S_2 be the sum of first 4n terms of the same arithmetic progression. If $(S_2–S_1)$ is 1000, then the sum of the first 6n terms of the arithmetic progression is equal to: **[JEE(Main)Mar.2021]**

माना एक समांतर श्रेढ़ी के प्रथम 2n पदों का योगफल S_1 है। माना उसी समांतर श्रेढ़ी के प्रथम 4n पदों का योगफल S_2 है। यदि $(S_2–S_1)$ =1000 है, तो इस समांतर श्रेढ़ी के प्रथम 6n पदों का योग बराबर है : **[JEE(Main)Mar.2021]**

Ans. 3000

Sol. $S_1 = \frac{2n}{2}\left(2a + \left(2n-1\right)d\right) = n\left(2a + (2n-1)d\right)$

$S_2 = \frac{4n}{2}\left(2a + \left(4n-1\right)d\right) = 2n\left(2a + (4n-1)d\right)$

$S_2 - S_1 = 1000$

$2n\,(2a + (4n-1)d) - n\,(2a + (2n-1)d) = 1000$

$n\,[4a + (4n-1)2d - 2a - (2n-1)d] = 1000$

$n\,[2a + d\,(8n - 2 - 2n + 1)] = 1000$

$n\,[2a + d\,(6n-1)] = 1000$

$\frac{6n}{2}\left(2a + \left(6n-1\right)d\right) = 3000$

$S = 3000$

Q.20 The total number of 4-digit numbers whose greatest common divisor with 18 is 3, is _____.

[JEE(Main)Feb.2021]

4-अंकों की संख्याओं, जिनका 18 के साथ महत्तम सर्वनिष्ठ भाजक 3 है, की कुल संख्या है _____

[JEE(Main)Feb.2021]

Ans. 1000

Sol. GCD (N, 18) = 3

Number must be an odd number and multiple of 3 and not a multiple of 9.

4-digit odd numbers multiple of 3 are :

1005, 1011, ..., 9999

Total such numbers are = 1500

4-digit odd number multiple of 9 are :

1017, 1035, ..., 9999

Total such numbers = 500

Required numbers = 1000.

2.4 ARITHMETIC MEAN

If three terms are in A.P. then the middle term is called the A.M. between the other two, so if a, b, c are in A.P., b is A.M. of a & c.

So A.M. of a and c = $\frac{a+c}{2} = b$.

A.M. of any n positive numbers $a_1, a_2 \ldots\ldots a_n$ is $A = \frac{a_1 + a_2 + a_3 + \ldots\ldots a_n}{n}$

n AM's between two given numbers

If in between two numbers 'a' and 'b' we have to insert n AM $A_1, A_2, \ldots\ldots A_n$

then $a, A_1, A_2, A_3 \ldots A_n, b$ will be in A.P.

The series consist of (n+2) terms and the last term is b and first term is a.

$\Rightarrow$ a + (n + 2 – 1) d = b

$\Rightarrow$ $d = \frac{b-a}{n+1}$

$A_1 = a + d$, $A_2 = a + 2d$ ..... $A_n = a + nd = b - d$

Note: Sum of n AM's inserted between a and b is equal to n times the single AM between a and b

i.e. $\sum_{r=1}^{n} A_r = nA$ where $A = \frac{a+b}{2}$

Q.1 Insert 20 AM between 4 & 67.

Ans. {7, 10, 13, ..., 64}

Sol. $d = \frac{b-a}{n+1} \Rightarrow \frac{67-4}{20+1}$

$d = \frac{63}{21} \Rightarrow 3$

7, 10, ..., 64

Q.2 101 AMs are inserted between 1 & 99. Find their sum.

Ans 5050

Sol. $d = \frac{99-1}{102} \Rightarrow \frac{98}{102} \Rightarrow \frac{49}{51}$

$A_1 = 1 + \frac{49}{51} \Rightarrow \frac{100}{51}$

$\text{Sum A.M.} = \frac{101}{2}\left[2\left(\frac{100}{51}\right) + (101-1)\frac{49}{51}\right]$

$= 101\left[\frac{100}{51} + \frac{50 \times 49}{51}\right]$

= 101 × 50

= 5050.

Q.3 If n AMs are inserted between 5 & 41 such that $\frac{A_3}{A_{n-1}} = \frac{2}{5}$ then find n.

Ans 11

Sol. a = 5 b = 41

41 = 5 + (n + 1)d

$\frac{a+3d}{a+(n-1)d} = \frac{2}{5}$ $\frac{36}{n+1} = d$

$$\frac{5+3d}{5+(n-1)d}=\frac{2}{5}$$

$25 + 15d = 10 + 2d\,(n - 1)$

$15 = d(2(n - 1) - 15)$

$$15=\frac{36}{n+1}(2n-2-15)$$

$5n + 5 = 12\,(2n - 17)$

$5n + 5 = 24n - 204$

$209 = 19n$

$11 = n$

3.1 GEOMETRIC PROGRESSION

Geometric Progression is defined as a series in which ratio between any two consecutive terms is constant throughout the series. This constant ratio is called as a common ratio.

$$r=\frac{T_2}{T_1}=\frac{T_3}{T_2}=\frac{T_4}{T_3}=.....$$

Standard Form

$a, ar, ar^2, ... ar^{n-1}$

'a' is the first term and 'r' is the common ratio

General Term of a G.P. :

$T_n = ar^{n-1}$

3.2 SUM OF FIRST 'n' TERMS OF A GP

$$S_n = a + ar + ar^2 + ... + ar^{n-1}$$

$$rS_n = \quad ar + ar^2 + ... \quad + ar^{n-1} + ar^n$$

$S_n - rS_n = a - ar^n$

$S_n(1-r) = a\,(1 - r^n)$

$$S_n = \frac{a\left(1-r^n\right)}{1-r} = \frac{a(r^n-1)}{r-1}\ \left(r\neq 1\right)$$

$S_n = nr$ when $r = 1$

Sum of an infinite G.P.

The sum of an infinite G.P. with first term a and common ratio r ($-1 < r < 1$ i.e. $|r| < 1$) is

$$S_\infty = \frac{a}{1-r}$$

3.3 HIGHLIGHTS OF A GP

1. No term of a GP can be zero
2. If a, b, c are in G.P. then $b^2 = ac$ but converse need not be true

3. If in a G.P, the product of two terms which are equidistant from the first and the last term, is constant and is equal to the product of first and last term. $\Rightarrow T_k . T_{n-k+1} = \text{constant} = a.\ell$

4. Three consecutive terms of a GP : $a/r,\ a,\ ar$;

   Five consecutive terms of a GP : $a/r^2,\ a/r,\ a,\ ar,\ ar^2$

5. If each term of a G.P. be multiplied or divided by the some non-zero quantity, then the resulting sequence is also a G.P.

6. If each term of a G.P. be raised to the same power, then resulting sequence is also a G.P.

7. If $a_1, a_2, a_3 ..... a_n$ is a G.P. of positive terms, then $\log a_1, \log a_2, ... \log a_n$ is an A.P. and vice-versa.

8. If $a_1, a_2, a_3 .....$ and $b_1, b_2, b_3 .....$ are two G.P.'s then $a_1b_1, a_2b_2, a_3b_3 .....$ & $\frac{a_1}{b_1}, \frac{a_2}{b_2}, \frac{a_3}{b_3} .....$ is also in G.P.

Q.1 For a geometric progression if $T_3 = 2, T_6 = -\frac{1}{4}$ find T_{10}.

Ans $-\frac{1}{64}$

Sol $ar^2 = 2$

$ar^5 = \frac{-1}{4}$

$\frac{1}{r^3} = -8 \Rightarrow r = \frac{-1}{2}$, $a = 8$

$T_{10} = 8\left(\frac{-1}{2}\right)^9 = -\frac{2^3}{2^9} = -\frac{1}{2^6} = -\frac{1}{64}$

Q.2 For a geometric progression if $T_1 + T_2 + T_3 = 19, T_1.T_2.T_3 = 216$ find S_∞.

Ans 27

Sol $T_1 = \frac{a}{r}, \quad T_2 = a, \quad T_3 = ar$

$T_1 . T_2 . T_3 = a^3 = 216$

$a = 6$

$\frac{6}{r} + 6 + 6r = 19$

$6 + 6r + 6r^2 = 19r$

$6r^2 - 13r + 6 = 0$

$6r^2 - 9r - 4r + 6 = 0$

$3r(2r - 3) - 2(2r - 3) = 0$

$(2r - 3)(3r - 2) = 0$

$r = \frac{2}{3}, \frac{3}{2}$ (rejected) $\quad T_1 = \frac{6}{2} \times 3 = 9$

$$S_\infty = \frac{9}{1-\frac{2}{3}} = 27$$

Q.3 If sum of an infinite GP is 15 & sum of their squares is 45 find G.P.

Ans $5, \frac{10}{3}, \frac{20}{9} \ldots$

Sol $S_\infty = \frac{a}{1-r} = 15$ .....(1)

a, ar, ar^2, ...

$a^2 + a^2r^2 + a^2r^4 + \ldots\ldots\ldots\infty = 45$

$\frac{a^2}{1-r^2} = 45$ .....(2)

from (1) and (2)

r = 2/3, a = 5

GP $\Rightarrow 5, \frac{10}{3}, \frac{20}{9}$

Q.4 For a geometric progression if $T_1 + T_2 + T_3 = 13,\ T_1^2 + T_2^2 + T_3^2 = 91$ find G.P.

Ans 1, 3, 9, ... or 9, 3, 1, ...

Sol $a + ar + ar^2 = 13$

$\frac{a(r^3-1)}{r-1} = 13 \quad \Rightarrow \quad a(r^2 + r + 1) = 13$

$a^2(r^2 + r + 1)^2 = 169$ .....(1)

$a^2 + a^2r^2 + a^2r^4 = 91$

$\frac{a^2(r^6-1)}{r^2-1} = 91$

$a^2(r^4 + r^2 + 1) = 91$ .....(2)

$\frac{(r^2+r+1)(r^2+r+1)}{(r^2-r+1)(r^2+r+1)} = \frac{169}{91} = \frac{13}{7}$

$7r^2 + 7r + 7 = 13r^2 - 13r + 13$

$6r^2 - 20r + 6 = 0$

$6r^2 - 18r - 2(r-3) = 0$

$(6r - 2)(r - 3) = 0$

r = 3, $\frac{1}{3}$

a = 1, 9

GP = 1, 3, 9, ... or 9, 3, 1, ...

Q.5 For a geometric progression($T_1 > 0$) if $T_1 - T_2 = 35$, $T_3 - T_4 = 560$ find G.P.

Ans 7, –28, 112, ...

Sol $a - ar = 35$ $\qquad ar^2 - ar^3 = 560$

$a(1 - r) = 35$ $\qquad ar^2 (1 - r) = 560$

$$\frac{1}{r^2} = \frac{35}{560} = \frac{1}{16}$$

$r = \pm 4$

$r = 4$ $\qquad a(1 - 4) = 35$ $\qquad r = -4$ $\qquad a(5) = 35$

$$a = \frac{-35}{3}$$

GP = 7, –28, 112, ... or $\frac{-35}{3}, \frac{140}{3}, \frac{-560}{3}$.....

Q.6 For a geometric progression if $T_p = x$, $T_q = y$, $T_r = z$, prove that $x^{q-r} y^{r-p} z^{p-q} = 1$.

Sol $ab^{p-1} = x$

$ab^{q-1} = y$

$ab^{r-1} = z$

$(ab^{p-1})^{q-r} (ab^{q-1})^{r-p} (ab^{r-1})^{p-q}$

$a^{q-r}b^{pq-pr-p+r}a^{r-p}b^{qr-qp-r+p}a^{p-q}b^{rp-rq-p+q}$

$a^o b^o = 1$ Hence proved.

Q.7 If p^{th}, q^{th}, r^{th} & s^{th} term of an AP are in G.P. then prove that (p – q), (q – r), (r – s) will b in G.P.

Sol
$$\left.\begin{aligned} A + (p-1)d &= a \\ A + (q-1)d &= b \\ A + (r-1)d &= c \\ A + (s-1)d &= e \end{aligned}\right\} \longrightarrow \text{GP}$$

So, b – a, c – b, e – c → GP

$$\frac{c-b}{b-a} = \frac{e-c}{c-b}$$

$$\frac{r-q}{q-p} = \frac{s-r}{r-q}$$

$$\frac{q-r}{p-q} = \frac{r-s}{q-r}$$ Hence proved

Q.8 If $a_1, a_2 \ldots\ldots\ldots\ldots\ldots a_n$ are in G.P. then prove that their product = $(a_1 . a_n)^{n/2}$.

Ans.

Sol. $a_1, a_2 \ldots\ldots\ldots\ldots a_n \rightarrow$ G.P

$a, ar, ar^2, ar^3 \ldots\ldots\ldots\ldots ar^{n-1}$

$a.\ ar^1.\ ar^2.\ ar^3.\ .....\ .....\ ar^{n-1}$

$a^n.\ r^{(1+2+3.....n-1)}$

$a^n \cdot r^{\frac{(n-1)(n)}{2}}$

$a^{n/2} a^{n/2} r^{\frac{(n-1)(n)}{2}}$

$\left(a \cdot ar^{n-1}\right)^{n/2}$

$\left(a_1 \cdot a_n\right)^{n/2}$

Q.9 The sum of first four terms of a geometric progression (G.P.) is $\frac{65}{12}$ and the sum of their respective reciprocals is $\frac{65}{18}$. If the product of first three terms of the G.P. is 1, and the third term is α, then 2α is ___.

[JEE(Main)Feb.2021]

एक गुणोत्तर श्रेढ़ी के पहले चार पदों का योग $\frac{65}{12}$ है तथा उनके व्युत्क्रमों का योग $\frac{65}{18}$ है। यदि इसके पहले तीन पदों का गुणनफल 1 हो और तीसरा पद α हो, तो 2α बराबर है _____ **[JEE(Main)Feb.2021]**

Ans. 3

Sol. GP a, ar, ar^2, ar^3

$a + ar + ar^2 + ar^3 = \frac{65}{12}$

$S_4 = a\frac{\left(r^4-1\right)}{r-1} = \frac{65}{12}$ .....(1)

$\frac{1}{a} + \frac{1}{ar} + \frac{1}{ar^2} + \frac{1}{ar^3} = \frac{65}{18}$

$\frac{\frac{1}{ar^3}\left(r^4-1\right)}{r-1} = \frac{65}{18}$ .....(2)

$\frac{(1)}{(2)}$

$a^2r^3 = \frac{3}{2}$

$a \times ar \times ar^2 = 1$

$a^3r^3 = 1$

$\Rightarrow a = \frac{2}{3}$

$r = \frac{3}{2}$

$\alpha = ar^2 = \frac{3}{2}$

$2\alpha = 3$

Q.10 In an increasing geometric series, the sum of the second and the sixth term is $\frac{25}{2}$ and the product of the third and fifth term is 25. Then, the sum of 4th, 6th and 8th terms is equal to : **[JEE(Main)Feb.2021]**

एक वर्धमान गुणोत्तर श्रेणी में, दूसरे तथा छठे पदों का योगफल $\frac{25}{2}$ है तीसरे तथा पाँचवें पदों का गुणनफल 25 है, तो चौथे, छठे तथा आठवें पदों का योगफल है : **[JEE(Main)Feb.2021]**

(A) 35 (B) 30 (C) 32 (D) 26

Ans. A

Sol. Let first term = a and common ratio = r.

Given $ar + ar^5 = \frac{25}{2}$ .....(1)

and $ar^2 . ar^4 = 25 \Rightarrow ar^3 = 5$ .....(2)

From (1) & (2) ; $r = \sqrt{2}$ and $a = \frac{5}{2\sqrt{2}}$.

$t_4 + t_6 + t_8 = ar^3 + ar^5 + ar^7$

$= ar^3 (1 + r^2 + r^4)$

$= 5 (1 + 2 + 4) = 35$

Sequences convertible to G.P.

Q.11 $S = 0.\overline{4} = 0.444 .....\infty$.

Sol $S = 0.4 + 0.04 + 0.004 + ...$

$= \frac{4}{10} + \frac{4}{10^2} + \frac{4}{10^3} + ...$

$S = \frac{\frac{4}{10}}{1 - \frac{1}{10}} = \frac{4}{9}$

Q.12 $S = 9 + 99 + 999 + ...n \text{ terms}$

Sol $S = (10 - 1) + (10^2 - 1) + (10^3 - 1) + ...+(10^n - 1)$

$S = (10 + 10^2 + 10^3 + ...+ 10^n) - n$

$$S = \frac{10(10^n - 1)}{9} - n$$

Q.13 $S = 4 + 44 + 444 + ...n \text{ terms}$

Sol $\frac{9}{4}S = 9 + 99 + 999 + ... \text{ n terms}$

$$\frac{9}{4}S = \frac{10(10^n - 1)}{9} - n = \frac{10(10^n - 1) - 9n}{9}$$

$$S = \frac{4}{9}\left[\frac{10(10^n - 1) - 9n}{9}\right]$$

Q.14 $S = 0.9 + 0.99 + 0.999 + ...n \text{ terms}$

Sol $S = \frac{9}{10} + \frac{99}{100} + \frac{999}{1000} + ... \text{ n terms}$

$$S = \frac{10-1}{10} + \frac{10^2-1}{100} + \frac{10^3-1}{1000} + ... \text{ n terms}$$

$$= \left(1 - \frac{1}{10}\right) + \left(1 - \frac{1}{10^2}\right) + \left(1 - \frac{1}{10^3}\right) + ... \text{ n terms}$$

$$n - \left[\frac{1}{10} + \frac{1}{10^2} + \frac{1}{10^3} + ...n \text{ terms}\right]$$

$$S = n - \left[\frac{\frac{1}{10}\left(1 - \frac{1}{10^n}\right)}{1 - \frac{1}{10}}\right] = n - \left(\frac{10^n - 1}{9.10^n}\right)$$

Q.15 $S = 0.2 + 0.22 + 0.\ 222 + ...n \text{ terms}$

Sol $S = \frac{2}{10} + \frac{22}{10^2} + \frac{222}{10^3} + ... \text{ n terms}$

$$\frac{9S}{2} = \frac{9}{10} + \frac{99}{10^2} + \frac{999}{10^3} + ... \text{ n terms}$$

$$\frac{9}{2}S = n - \left(\frac{10^n - 1}{9 \times 10^n}\right)$$

$$S = \frac{2}{9}\left[n - \left(\frac{10^n - 1}{9 \times 10^n}\right)\right]$$

3.4 GEOMETRIC MEAN

If a, b, c are in G.P., b is called as the G.M. of a & c.

If a and c are both positive, then $b = \sqrt{ac}$ and if a and c are both negative, then $b = -\sqrt{ac}$.

$b^2 = ac$, therefore $b = \sqrt{ac}$; $a > 0$, $c > 0$.

$GM(x, y) = \sqrt{xy}$

$GM(x, y, z) = (xyz)^{1/3}$

$GM(x_1, x_2, ...x_n) = (x_1 x_2 ...x_n)^{1/n}$

n GM's between two given numbers

If in between two numbers 'a' and 'b', we have to insert n GM $G_1, G_2....G_n$

then a, G_1, G_2 , ... G_n, b will be in GP.

The series consist of (n+2) terms and the last term is b and first term is a.

$\Rightarrow$ $ar^{n+2-1} = b$

$\Rightarrow$ $r = \left(\frac{b}{a}\right)^{\frac{1}{n+1}}$

$G_1 = ar$, $G_2 = ar^2$ ..... $G_n = ar^n = b/r$

Note : Product of n GM's inserted between 'a' and 'b' is equal to n^{th} power of the single GM between 'a' and 'b' i.e. $\prod_{r=1}^{n} G_r = (G)^n$ where $G = \sqrt{ab}$

Q.1 Insert 4 GMs between 5 & 160.

Ans. {10, 20, 40, 80}

Sol. $\left(\frac{160}{5}\right)^{1/5} = r$

$(32)^{\frac{1}{5}} = r$

$2 = r$

$a = 5$ $r = 2$

10, 20, 40, 80

Q.2 AM of 2 numbers x & y (x > y) is 15 & their GM is 9 find numbers.

Ans. $x = 27$, $y = 3$

Sol. $\frac{x + y}{2} = 15$ $\sqrt{xy} = 9$

$x + y = 30$ $xy = 81$

$t^2 - 30t + 81 = 0$ $\begin{cases} \nearrow t_1 = x \\ \searrow t_2 = y \end{cases}$

$(t - 27)(t - 3) = 0$

$t = 27$	or	$t = 3$
$x = 27$		$x = 3$
$y = 3$		$y = 27$

Q.3 If sum of two positive numbers a & b (a > b) is n times their GM then prove that $\frac{a}{b} = \frac{n + \sqrt{n^2 - 4}}{n - \sqrt{n^2 - 4}}$.

Sol $a + b = n\sqrt{ab}$

$a^2 + b^2 + 2ab = n^2ab$

$(a - b)^2 = (n^2 - 4)\, ab$

$$\frac{a^2 + b^2 + 2ab}{a^2 + b^2 - 2ab} = \frac{n^2}{n^2 - 4}$$

$$\frac{a + b}{a - b} = \frac{n}{\sqrt{n^2 - 4}}$$

Componendo and Dividendo

$$\frac{a}{b} = \frac{n + \sqrt{n^2 - 4}}{n - \sqrt{n^2 - 4}}$$ Hence proved

3.5 CONCEPT OF AM-GM

For two positive numbers a and b, AM ≥ GM always

$$\frac{a + b}{2} \geq \sqrt{ab}$$

For n positive numbers

AM ≥ GM

$$\frac{x_1 + x_2 + ...x_n}{n} \geq \left(x_1 x_2 .....x_n\right)^{\frac{1}{n}}$$

If AM = GM $\Rightarrow$ numbers are equal

Ex.1 $x^3 - 12x^2 + kx - 64 = 0$ If roots are positive find k

Sol roots are α, β and γ

$$AM = \frac{\alpha + \beta + \gamma}{3} = \frac{12}{3} = 4$$

$$GM = \left(\alpha\beta\gamma\right)^{\frac{1}{3}} = \left(64\right)^{1/3} = 4$$

$\because$ AM = GM

$\therefore\ \alpha = \beta = \gamma = 4$

Type-1: To minimize sum

Q.1 $\left(x+\frac{1}{x}\right)_{min} = ?$ $\quad x > 0$

Ans. 2

Sol. $\frac{x+\frac{1}{x}}{2} \geq \sqrt{x \times \frac{1}{x}}$

$x+\frac{1}{x} \geq 2$

$\left(x+\frac{1}{x}\right)_{min} = 2$

Q.2 $\left(\frac{a^2}{b}+\frac{3b}{a}+\frac{9}{a}\right)_{min} = ?$ $\quad a, b > 0$

Ans. 9

Sol. $\frac{\frac{a^2}{b}+\frac{3b}{a}+\frac{9}{a}}{3} \geq \sqrt[3]{\frac{a^2}{b}\cdot\frac{3b}{a}\cdot\frac{9}{a}}$

$\frac{a^2}{b}+\frac{3b}{a}+\frac{9}{a} \geq 9$

$\left.\frac{a^2}{b}+\frac{3b}{a}+\frac{9}{a}\right|_{min} = 9$

Q.3 $\left(x^2+\frac{16}{x}\right)_{min} = ?$ $\quad x > 0$

Ans. 12

Sol. $\frac{x^2+\frac{8}{x}+\frac{8}{x}}{3} \geq \sqrt[3]{x^2\cdot\frac{8}{x}\cdot\frac{8}{x}}$

$x^2+\frac{16}{x} \geq 12$

$\left(x^2+\frac{16}{x}\right)_{mini} = 12$

Type-2: To maximize product

Q.4 If $x + y = 12$ x, y > 0 Find $xy|_{max}$

Ans. 36

Sol. $\frac{x+y}{2} \geq \sqrt{xy}$

$6 \geq \sqrt{xy}$

$36 \geq xy$

$xy)_{max} = 36.$

Q.5 If $a + b + c = 15$ a, b, c > 0 Find $abc|_{max}$

Ans. 125

Sol. $\frac{a+b+c}{3} \geq \sqrt[3]{abc}$

$5 \geq \sqrt[3]{abc}$

$125 \geq abc$

$abc)_{max} = 125.$

Q.6 If $x + 2y + z = 16$ x, y, z > 0 Find (i) $xy^2z|_{max}$ (ii) $x^2yz|_{max}$

Ans. (i) 256 (ii) 512

Sol. (i) $xy^2z|_{max}$

$\frac{x+y+y+z}{4} \geq \sqrt[4]{x \cdot y^2 \cdot z}$

$4 \geq \sqrt[4]{xy^2z}$

$4^4 \geq xy^2z$

$xy^2z)_{max} = 256.$

(ii) x^2yz

$\frac{\frac{x}{2}+\frac{x}{2}+2y+z}{4} \geq \sqrt[4]{\frac{x}{2} \cdot \frac{x}{2} \cdot 2y \cdot z}$

$4 \geq \frac{1}{\sqrt[4]{2}} \sqrt[4]{x^2y \cdot z}$

$4\sqrt[4]{2} \geq \sqrt[4]{x^2y \cdot z}$

$4^4 \cdot 2 \geq x^2y \cdot z$

$2^9 \geq x^2y \cdot z$

$512 \geq x^2yz$

$x^2y)_{max} = 512.$

Q.7 If, $a + b + c = 7$ $a, b, c > 0$ find $a^2b^3c^2\big|_{max}$.

Ans. 432

Sol. $$\frac{\frac{a}{2}+\frac{a}{2}+\frac{b}{3}+\frac{b}{3}+\frac{b}{3}+\frac{c}{2}+\frac{c}{2}}{7} \geq \sqrt[7]{\frac{a^2}{4}\cdot\frac{b^3}{27}\cdot\frac{c^2}{4}}$$

$4.27.4. \geq a^2 . b^3 . c^2.$

$432 \geq a^2.b^3.c^2$

$a^2.b^3.c^2)_{max} = 432$

Q.8 If $a + b + c = 7$, $a, b, c > 0$, $a^2 b^3 c^2 = 432$ find a, b, c.

Ans. $a = 2, b = 3, c = 2$

Sol. From above we know.

$a^2 b^3 c^2 = 4.27.4.$

$a = 2 \quad b = 3 \quad c = 2$

Q.9 Prove that $(x + y)(y + z)(z + x) \geq 8xyz$, $x, y, z > 0$.

Sol. $x + y \geq 2\sqrt{xy}$

$y + z \geq 2\sqrt{yz}$

$z + x \geq 2\sqrt{zx}$

$(x + y)(y + z)(z + x) \geq 8\sqrt{x^2y^2z^2}$

$(x + y)(y + z)(z + x) \geq 8xyz$

Q.10 Prove that ΔABC is equilateral triangle if and only if $\tan A + \tan B + \tan C = 3\sqrt{3}$.

Sol Case-1 : Let ABC is equilateral then $\tan A + \tan B + \tan C = 3\sqrt{3}$ is obviously true

Case-2 : Let $\tan A + \tan B + \tan C = 3\sqrt{3}$ be true

$$\frac{\tan A + \tan B + \tan C}{3} = \sqrt{3}$$

AM of tanA, tanB, tanC = $\sqrt{3}$

now GM of tanA, tanB, tanC = $(\tan A \tan B \tan C)^{1/3} = \left(3\sqrt{3}\right)^{1/3} = \left(3^{3/2}\right)^{1/3} = \sqrt{3}$

Hence AM = GM ; $\therefore$ tanA = tan B = tan C $\Rightarrow$ Δ is equilateral

4.1 HARMONIC PROGRESSION

A sequence is said to be in H.P if the reciprocals of its terms are in A.P..
If the sequence $a_1, a_2, a_3, \ldots, a_n$ is in H.P. then $1/a_1, 1/a_2, \ldots, 1/a_n$ is in A.P.

Standard form

$$\frac{1}{a}, \frac{1}{a+d}, \frac{1}{a+2d} + \ldots\ldots\ldots\ldots$$

If a, b, c are in HP

$\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are in AP

$$2\left(\frac{1}{b}\right) = \frac{1}{a} + \frac{1}{c}$$

$$\frac{2}{b} = \frac{a+c}{ac} \Rightarrow \quad b = \frac{2ac}{a+c}$$

Note :

(i) There is no general formula for S_n of an HP.

(ii) Problems based on HP are generally solved by converting them into carresponding AP.

(iii) For every HP, there exist a corresponding AP but converse need not be true.

Q.1 For a harmonic progression if $T_5 = 4$, $T_4 = 5$ find T_{10}?

Ans. 2

Sol. $T_5 = \frac{1}{4} \quad T_4 = \frac{1}{5}$

$$a + 4d = \frac{1}{4}$$

$$a + 3d = \frac{1}{5}$$

$$- \quad - \quad -$$

$$d = \frac{1}{4} - \frac{1}{5} \Rightarrow \frac{1}{20}$$

$$a = \frac{1}{4} - \frac{1}{5} \Rightarrow \frac{1}{20}$$

$T_{10} = a + 9d$

$= \frac{1}{20} + \frac{9}{20} \Rightarrow \frac{10}{20} = \frac{1}{2}$

in H.P. $T_{10} = 2$.

Q.2 For a harmonic progression if $T_3 = \frac{1}{3}$, $T_6 = \frac{1}{5}$, $T_n = \frac{3}{203}$, find n.

Ans 100

Sol. A.P. $T_3 = 3$ $T_6 = 5$ $T_n = \frac{203}{3}$

$a + 5d = 5$

$a + 2d = 3$

$- \quad - \quad -$

$3d = 2$

$d = \frac{2}{3}$

$a = 3 - \frac{4}{3} \Rightarrow \frac{5}{3}$

$a + (n-1)d = \frac{203}{3}$

$\frac{5}{3} + (n-1)\frac{2}{3} = \frac{203}{3}$

$n - 1 = \frac{203 - 5}{2}$

$n = 100$

Q.3 If a^2, b^2, c^2, are in AP.

(i) Prove that b + c, c + a, a +b are in HP. (ii) Prove that $\frac{b+c}{a}, \frac{c+a}{b}, \frac{a+b}{c}$ are in HP.

Sol. (i) $a^2, b^2, c^2 \rightarrow$ AP

$\frac{1}{b+c}, \frac{1}{c+a}, \frac{1}{a+b} \rightarrow A \cdot P.$

Multiply by (a + b) (b + c) (c + a)

(a + b) (c + a), (a + b) (b + c), (b + c) (c + a)

Subtract (ab + bc + ca)

$a^2, b^2, c^2 \rightarrow$ AP

$\therefore$ (b + c), (c + a), (a + b) $\rightarrow$ HP

(ii) $\frac{a}{b+c}, \frac{b}{c+a}, \frac{c}{a+b} \rightarrow A \cdot P.$

Add, $\frac{a+b+c}{b+c}, \frac{a+b+c}{c+a}, \frac{a+b+c}{a+b}$

Divide by a + b + c

$\frac{1}{b+c}, \frac{1}{c+a}, \frac{1}{a+b} \rightarrow A \cdot P.$

Prove above

Q.4 If $a_1, a_2, ...a_n$ are in HP. Pove that $a_1a_2 + a_2a_3 + ...+ a_{n-1}\, a_n = (n-1)a_1a_n$.

Sol $\frac{1}{a_1}, \frac{1}{a_2}, \frac{1}{a_3} ..... \frac{1}{a_n}$ are in A.P.

$$d = \frac{a_1 - a_2}{a_1a_2} = \frac{a_2 - a_3}{a_2a_3} = ... \frac{a_{n-1} - a_n}{a_na_{n-1}}$$

$$\text{now LHS} = \frac{1}{d}[(a_1 - a_2) + (a_2 - a_3) + ... + (a_{n-1} - a_n)] = \frac{a_1 - a_n}{d}$$

$$\text{now} \quad \frac{1}{a_n} = \frac{1}{a_1} + (n-1)d \quad \Rightarrow \quad (n-1)d = \frac{a_1 - a_n}{a_na_1}$$

$$d = \frac{a_1 - a_n}{(a_na_1)(n-1)} \quad ; \quad \therefore \quad \text{LHS} = (n-1)a_1a_n = \text{RHS}$$

Q.5 If a, b, c are in AP; b, c, a are in GP then prove that c, a, b will be in HP.

Sol. a, b, c $\rightarrow$ AP b, c, a $\rightarrow$ GP

$a + c = 2b$ $ab = c^2$

$$\frac{a+c}{b} = 2$$

$$\frac{a+c}{ab} = \frac{2}{a}$$

$$\frac{a}{b} + \frac{c}{ab} = \frac{2}{a}$$

$$\frac{1}{b} + \frac{c}{c^2} = \frac{2}{a}$$

$$\frac{1}{b} + \frac{1}{c} = \frac{2}{a}$$

Q.6 $\frac{a-x}{px}=\frac{a-y}{qy}=\frac{a-z}{rz}$. If p, q, r are in AP. Prove that x, y, z are in HP.

Sol. $\frac{a-x}{px}=\frac{a-y}{qy}=\frac{a-z}{rz}=K$ $\quad 2q = p + r$

$p=\frac{a-x}{kx}$ $\quad q=\frac{a-y}{ky}$ $\quad r=\frac{a-z}{kz}$

$2\left(\frac{a-y}{ky}\right)=\frac{a-x}{kx}+\frac{a-z}{kz}$

$2\left(\frac{a}{y}-1\right)=\frac{a}{x}-1+\frac{a}{z}-1$

$2\frac{a}{y}-2=\frac{a}{x}-1+\frac{a}{z}-1$

$\frac{2}{y}=\frac{1}{x}+\frac{1}{z}$

x.y.z. $\rightarrow$ H.P.

Q.7 $a^x = b^y = c^z = d^w$. If a, b, c, d are in GP. Prove that x, y, z, w are in HP.

Sol If $a^x = b^y = c^z = d^w = ... = k$

$a = k^{\frac{1}{x}}$; $b = k^{\frac{1}{y}}$; $c = k^{\frac{1}{z}}$; $d = k^{\frac{1}{w}}$

$\therefore$ a, b, c, d are in G.P. $\Rightarrow$ $\frac{b}{a}=\frac{c}{b}=\frac{d}{c}$

i.e. $\frac{k^{1/y}}{k^{1/x}}=\frac{k^{1/z}}{k^{1/y}}=\frac{k^{1/w}}{k^{1/z}}=.....$

$k^{\frac{1}{y}-\frac{1}{z}}=k^{\frac{1}{z}-\frac{1}{y}}=k^{\frac{1}{w}-\frac{1}{z}}$

$\frac{1}{y}-\frac{1}{x}=\frac{1}{z}-\frac{1}{y}=\frac{1}{w}-\frac{1}{z}$

$\frac{1}{x}, \frac{1}{y}, \frac{1}{z}, \frac{1}{w}$, ... are in A.P. $\Rightarrow$ x, y, z, w ..... H.P.]

4.2 HARMONIC MEAN

If three or more than three terms are in H.P., then all the numbers lying between first and last term are called Harmonical Means between them. i.e.

The H.M. between two given quantities a and b is H so that a, H , b are in H.P.

i.e. $\frac{1}{a}, \frac{1}{H}, \frac{1}{b}$ are in A.P.

$\frac{1}{H}-\frac{1}{a}=\frac{1}{b}-\frac{1}{H} \Rightarrow H=\frac{2ab}{a+b}$

HM of numbers is reciprocal of arithmetic mean of reciprocal of numbers

$HM(x, y)=\frac{2}{\frac{1}{x}+\frac{1}{y}}$

$HM(x, y, z)=\frac{3}{\frac{1}{x}+\frac{1}{y}+\frac{1}{z}}$

$HM(x_1, x_2, ...x_n)=\frac{n}{\frac{1}{x_1}+\frac{1}{x_2}+.....\frac{1}{x_n}}$

Insertion of 'n' HM's between a and b :

Let $H_1, H_2.....H_n$ are n HM's between a and b

hence a, $H_1, H_2, ...H_n$ b are in H.P.

$\frac{1}{a}, \frac{1}{H_1}, \frac{1}{H_2}.....\frac{1}{H_n}, \frac{1}{b}$ are in A.P.

$\frac{1}{b}=\frac{1}{a}+(n+1)d \quad ; \frac{1}{b}-\frac{1}{a}=(n+1)d \; ; \; d=\frac{a-b}{ab(n+1)}$

$\frac{1}{H_1}=\frac{1}{a}+d, \; \frac{1}{H_2}=\frac{1}{a}+2d, \; \frac{1}{H_3}=\frac{1}{a}+3d .... \quad \frac{1}{H_n}=\frac{1}{a}+nd$

Note : Sum of reciprocals of n HMs between two numbers is equal to n times reciprocal of their single HM.

$$\frac{1}{H_1}+\frac{1}{H_2}+.....+\frac{1}{H_n}=n\left(\frac{\frac{1}{a}+\frac{1}{b}}{2}\right)=n\left(\frac{1}{H}\right)$$

5. RELATION BETWEEN AM, GM AND HM

(i) If A, G, H, are respectively A.M., G.M., H.M. between two positive number a & b then

(a) $G^2 = AH$ (A, G, H constitute a GP) (b) $A \geq G \geq H$ (c) $A = G = H \Leftrightarrow a = b$

(ii) Let $a_1, a_2, ... , a_n$ be n positive real numbers, then we define their arithmetic mean (A), geometric mean (G) and harmonic mean (H) as $A=\frac{a_1+a_2+.....+a_n}{n}$

$G=(a_1 a_2.....a_n)^{1/n}$ and $H=\frac{n}{\left(\frac{1}{a_1}+\frac{1}{a_2}+\frac{1}{a_3}+....\frac{1}{a_n}\right)}$

$AM \geq GM \geq HM$.

Moreover equality holds at either place if and only if $a_1 = a_2 = \ldots\ldots = a_n$

$RMS \geq AM \geq GM \geq HM$

RMS = Root mean square

$$RMS\ (a, b) = \sqrt{\frac{a^2 + b^2}{2}}$$

$$RMS\ (a, b, c) = \sqrt{\frac{a^2 + b^2 + c^2}{3}}$$

Q.1 If a, b, c > 0 and a + b + c = 6 find $\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right)_{min.}$

Ans 3/2

Sol $AM \geq HM$

$$\frac{a+b+c}{3} \geq \frac{3}{\frac{1}{a} + \frac{1}{b} + \frac{1}{c}}$$

$$2 \geq \frac{3}{\frac{1}{a} + \frac{1}{b} + \frac{1}{c}}$$

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq \frac{3}{2}$$

Q.2 If a, b, c are in GP ; x is AM of a & b; y is AM of b & c.

(i) Prove that $\frac{1}{x} + \frac{1}{y} = \frac{2}{b}$. (ii) Prove that $\frac{a}{x} + \frac{c}{y} = 2$

Sol (i) $x = \frac{a+b}{2}, \quad y = \frac{b+c}{2}$

$$\frac{1}{x} + \frac{1}{y} = \frac{2}{a+b} + \frac{2}{b+c}$$

$$2\left(\frac{b+c+a+b}{ab+ac+b^2+bc}\right) = 2\left(\frac{2b+a+c}{ab+bc+2b^2}\right) = \frac{2}{b}$$

(ii) $a, ar, ar^2 \equiv (a, b, c)$

$$x = \frac{a+ar}{2}, y = \frac{ar+ar^2}{2}$$

$$\frac{2a}{a+ar} + \frac{2ar^2}{ar+ar^2}$$

$$\frac{2}{1+r}+\frac{2r}{1+r}=2\left(\frac{1+r}{1+r}\right)=2$$

Q.3 If a, b, c are three distinct positive reals in H.P. then prove that $a^n + c^n > 2b^n$.

[Sol. using AM > GM in a^n and c^n

$$\frac{a^n+c^n}{2} > (a^n c^n)^{1/2}$$

i.e. $a^n + c^n > 2(ac)^{n/2}$ ....(1)

a, b, c are in H.P. $\Rightarrow$ b is the HM between a and c.

hence applying GM > HM

$\sqrt{ac} > b$; $(ac)^{n/2} > b^n$; $2(ac)^{n/2} = 2b^n$ ....(2)

from (1) and (2)

$a^n + c^n > 2b^n$

6. ARITHMETIC-GEOMETRIC PROGRESSION(AGP)

A series, each term of which is formed by multiplying the corresponding term of an A.P. & G.P. is called the Arithmetic-Geometric Series , e.g. $1+ 3x + 5x^2 + 7x^3 + ...$

Here 1, 3, 5, ... are in A.P. & 1, x, x^2, x^3 ..... are in G.P.

Let $S_n = a + (a + d)r + (a + 2d)r^2 + ... + [a + (n-1)d]r^{n-1}$

then $S_n = \frac{a}{1-r} + \frac{dr(1-r^{n-1})}{(1-r)^2} - \frac{[a+(n-1)d]\ r^n}{1-r}$, $r \neq 1$

If $0 < |r| < 1$ & $n \to \infty$, then $\lim_{n\to\infty} r^n = 0$, $S_\infty = \frac{a}{1-r} + \frac{dr}{(1-r)^2}$

Find Sum S

1. $S = 1 + 2x + 3x^2 + ... + nx^{n-1}$

Ans. $\frac{1-x^n}{(1-x)^2} - \frac{nx^n}{1-x}$

Sol. $S = 1 + 2x + 3x^2 + ... + nx^{n-1}$

$xS = x + 2x^2 + 3x^3 ..... + (n-1)x^{n-1} + nx^n$

– – – – – –

$S(1-x) = 1 + x + x^2 + x^3 ..... x^{n-1} - nx^n$

$S(1-x) = \frac{1(1-x^n)}{1-x} - nx^n$

$S = \frac{1-x^n}{(1-x)^2} - \frac{nx^n}{1-x}$

2. $S = 1 + \frac{4}{5} + \frac{7}{5^2} + \frac{10}{5^3} + ...$n terms

Ans $\frac{35}{16} - \frac{(12n+7)}{16 \cdot 5^{n-1}}$.

Sol. $S = 1 + \frac{4}{5} + \frac{7}{5^2} + \frac{10}{5^3} + ... + \frac{3n-2}{5^{n-1}}$ .....(i)

$\left(\frac{1}{5}\right) S = \frac{1}{5} + \frac{4}{5^2} + \frac{7}{5^3} + ... + \frac{3n-5}{5^{n-1}} + \frac{3n-2}{5^n}$ .....(ii)

(i) – (ii) $\Rightarrow$

$\frac{4}{5} S = 1 + \frac{3}{5} + \frac{3}{5^2} + \frac{3}{5^3} + ... + \frac{3}{5^{n-1}} - \frac{3n-2}{5^n}$.

$\frac{4}{5} S = 1 + \frac{\frac{3}{5}\left(1-\left(\frac{1}{5}\right)^{n-1}\right)}{1-\frac{1}{5}} - \frac{3n-2}{5^n} = 1 + \frac{3}{4} - \frac{3}{4} \times \frac{1}{5^{n-1}} - \frac{3n-2}{5^n}$

$\frac{4}{5} S = \frac{7}{4} - \frac{12n+7}{4.5^n}$ $\therefore S = \frac{35}{16} - \frac{(12n+7)}{16 \cdot 5^{n-1}}$.

3. $S = 1 + 2x + 3x^2 + 4x^3 + ...\infty$

Ans $\frac{1}{(1-x)^2}$

Sol. $S = 1 + 2x + 3x^2 + 4x^3 + ...$ .....(i)

$xS = x + 2x^2 + 3x^3 + ...$ .....(ii)

(i) - (ii) $\Rightarrow$

$(1 - x) S = 1 + x + x^2 + x^3 + ...$

or $S = \frac{1}{(1-x)^2}$

4. $S = 1 + \frac{4}{5} + \frac{7}{5^2} + \frac{10}{5^3} + ...$n terms.$\infty$

Ans. $\frac{35}{16}$

Sol. $S = 1 + \frac{4}{5} + \frac{7}{5^2} + \frac{10}{5^3} + ... \infty$

$\frac{S}{5} = \frac{1}{5} + \frac{4}{5^2} + \frac{7}{5^3} + \cdots\cdots$

$\frac{4S}{5} = 1 + \frac{3}{5} + \frac{3}{5^2} + \frac{3}{5^3} \cdots\cdots \infty$

$$\frac{4}{5}S = 1 + \frac{\frac{3}{5}}{1-\frac{1}{5}}$$

$$\frac{4}{5}S = 1 + \frac{3}{4}$$

$$S = \frac{7}{4} \times \frac{5}{4} \Rightarrow \frac{35}{16}$$

5. $S = \frac{3}{5} + \frac{5}{15} + \frac{7}{45} + \frac{9}{135} + \ldots\ldots\ldots\infty$

Ans. $\frac{6}{5}$

Sol. $= \frac{3}{5} + \frac{5}{15} + \frac{7}{45} + \frac{9}{135} + \ldots\ldots\ldots\infty$

$$\frac{S}{3} = \frac{3}{15} + \frac{5}{45} + \frac{7}{135} + \cdots + \infty$$

$$\frac{2S}{3} = \frac{3}{5} + \frac{2}{15} + \frac{2}{45} + \frac{2}{135} - \cdots$$

$$\frac{2S}{3} = \frac{3}{5} + \frac{\frac{2}{15}}{1-\frac{1}{3}}$$

$$\frac{2S}{3} = \frac{3}{5} + \frac{1}{5}$$

$$S = \frac{6}{5}$$

6. $S = 1 + 3x + 6x^2 + 10x^3 + \ldots\ldots\ldots\infty$

Ans. $\frac{1}{(1-x)^3}$

Sol. $S = 1 + 3x + 6x^2 + 10x^3 + \ldots\ldots\ldots\infty$

$xS = x + 3x^2 + 6x^2 + \ldots\ldots\ldots\ldots\ldots$

$S(1 - x) = 1 + 2x + 3x^2 + 4x^3 + \ldots\ldots\ldots\infty$

$xS(1 - x) = x + 2x^2 + 3x^2 + \ldots\ldots\ldots$

$(1-x)^2 S = 1 + x + x^2 + x^3 + \ldots\ldots\ldots\infty$

$$(1-x)^2 S = \frac{1}{1-x}$$

$$S = \frac{1}{(1-x)^3}$$

7. $S = \frac{3}{19} + \frac{33}{19^2} + \frac{333}{19^3} + \frac{3333}{19^4} + \ldots\ldots\ldots\infty$

Ans. $\frac{19}{54}$

Sol. $$S = 3\left(\frac{1}{19} + \frac{11}{19^2} + \frac{111}{19^3} + \frac{1111}{19^4} + \cdots\infty\right)$$

$$= \frac{3}{9}\left(\frac{9}{19} + \frac{99}{19^2} + \frac{999}{19^3} + \frac{9999}{19^4}\ldots\infty\right)$$

$$= \frac{3}{9}\left(\frac{10-1}{19} + \frac{100-1}{19^2} + \frac{1000-1}{19^3} + \frac{10000-1}{19^4} + \ldots\infty\right)$$

$$= \frac{3}{9}\left[\left(\frac{10}{19} + \frac{100}{19^2} + \frac{1000}{19^3} + \ldots\infty\right) - \left(\frac{1}{19} + \frac{1}{19^2} + \frac{1}{19^3}\ldots\infty\right)\right]$$

$$= \frac{3}{9}\left[\left(\frac{\frac{10}{19}}{1-\frac{10}{19}}\right) - \left(\frac{\frac{1}{19}}{1-\frac{1}{19}}\right)\right]$$

$$= \frac{3}{9}\left[\frac{\frac{10}{19}}{\frac{9}{19}} - \left(\frac{\frac{1}{19}}{\frac{18}{19}}\right)\right]$$

$$= \frac{3}{9}\left[\frac{10}{9} - \frac{1}{18}\right]$$

$$= \frac{3}{9}\left[\frac{20-1}{18}\right]$$

$$= \frac{1}{3} \times \frac{19}{18} \Rightarrow \frac{19}{54}$$

7.1 MISCELLANEOUS SEQUENCES - TYPE 1

Sigma Notations (Σ)Theorems :

(a) $\sum_{r=1}^{n}(a_r \pm b_r) = \sum_{r=1}^{n} a_r \pm \sum_{r=1}^{n} b_r$

(b) $\sum_{r=1}^{n} k\,a_r = k\sum_{r=1}^{n} a_r$

(c) $\sum_{r=1}^{n} k = nk$; where k is a constant.

Some Important Results

(a) $\sum_{r=1}^{n} r = 1+2+3+...4 = \frac{n(n+1)}{2}$ (sum of the first n natural numbers)

(b) $\sum_{r=1}^{n} r^2 = 1^2+2^2+3^2+....n^2 = \frac{n(n+1)(2n+1)}{6}$ (sum of the squares of the first n natural numbers)

(c) $\sum_{r=1}^{n} r^3 = 1^3+2^3+3^3+...n^3 = \frac{n^2(n+1)^2}{4} = \left[\sum_{r=1}^{n} r\right]^2$ (sum of the cubes of the first n natural numbers)

(d) $\sum_{r=1}^{n} 1 = 1+1+1+...1 = n$

Note :

If n^{th} term of a sequence is given by $T_n = an^3 + bn^2 + cn + d$ where a, b, c, d are constants,

then sum of n terms $S_n = \Sigma T_n = a\Sigma n^3 + b\Sigma n^2 + c\Sigma n + \Sigma d$

1. If $T_r = 2r^2 + 3r + 2$ find S_n.

Ans. $\frac{2n(n+1)(2n+1)}{6} + \frac{3n(n+1)}{2} + 2n$

Sol. $\sum_{r=1}^{n} T_r = 2\sum_{r=1}^{n} r^2 + 3\sum_{r=1}^{n} r + \sum_{r=1}^{n} 2$

$= 2\frac{n(n+1)(2n+1)}{6} + 3\frac{n(n+1)}{2} + 2n$

$= \frac{n(n+1)(2n+1)}{3} + \frac{3n(n+1)}{2} + 2n$

2. If $T_r = r^2 + r + 3^{r-1}$ find S_n.

Ans. $\frac{n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2} + \frac{3^n - 1}{2}$

Sol. $\sum_{r=1}^{n} T_r = \sum_{r=1}^{n} r^2 + \sum_{r=1}^{n} r + \frac{1}{3}\sum_{r=1}^{n} 3^r$

$S_n = \frac{n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2} + \frac{1}{3}\frac{3\left(3^n - 1\right)}{2}$

3. If $T_r = (r + 2)(r + 1)$ find S_n.

Ans. $\frac{n(n+1)(2n+1)}{6} + \frac{3n(n+1)}{2} + 2n$

Sol. $T_r = r^2 + 3r + 2$

$$\sum_{r=1}^{n} T_r = \sum_{r=1}^{n} r^2 + 3\sum_{r=1}^{n} r + \sum_{r=1}^{n} 2$$

$$= \frac{n(n+1)(2n+1)}{6} + \frac{3n(n+1)}{2} + 2n$$

4. $31^2 + 32^2 + \ldots\ldots\ldots\ldots (50)^2 = ?$

Ans. 33470

Sol. $(1^2 + 2^2 + \ldots\ldots 50^2) - (1^2 + 2^2 + \ldots\ldots 30^2)$

$$\frac{50(51)(101)}{6} - \frac{30(31)(61)}{6}$$

$= 33470$

5. $(1 \times 2) + (2 \times 3) + (3 \times 4) + \ldots\ldots\ldots$find T_n & S_n.

Ans. $T_n = n(n+1)$ $\quad S_n = \frac{n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2}$

Sol. $T_n = n(n + 1)$

$= n^2 + n$

$$S_n = \frac{n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2}$$

6. $(2 \times 5) + (3 \times 7) + (4 \times 9) + \ldots\ldots\ldots$find T_n.

Ans. $T_n = (n+1)(2n+3)$

Sol. $(2 \times 5) + (3 \times 7) + (4 \times 9) + \ldots\ldots\ldots$

$T_n = [2 + (n - 1)] \times [5 + (n - 1)2]$

$= (n + 1)(2n + 3)$

7. Find $\sum_{i=1}^{n} \sum_{j=1}^{i} \sum_{k=1}^{j} (1)$

Ans $\frac{n(n+1)(n+2)}{6}$

Sol. $\sum_{i=1}^{n} \sum_{j=1}^{i} \sum_{k=1}^{j} 1 = \sum_{i=1}^{n} \sum_{j=1}^{i} j = \sum_{i=1}^{n} \frac{i(i+1)}{2} = \frac{1}{2}\left[\sum_{i=1}^{n} i^2 + \sum_{i=1}^{n} i\right] = \frac{1}{2}\left[\frac{n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2}\right]$

$= \frac{n(n+1)}{12}[2n+1+3] = \frac{n(n+1)(n+2)}{6}$.

8. Find nth term of the series $1 \cdot 3 \cdot 5 + 3 \cdot 5 \cdot 7 + 5 \cdot 7 \cdot 9 + ...$

Ans $T_n = (2n-1)(2n+1)(2n+3)$

Sol. If we take first iteration each terms then we will get.

1, 3, 5, ...

So r^{th} term of this sequence will be

$2r - 1$

Similarly for 3, 5, 7..... if will be $2r + 1$

for 5, 7, 9.....if will be $2r + 3$.

$\therefore T_n = (2n-1)(2n+1)(2n+3)$

9. $\frac{1^3}{1} + \frac{1^3+2^3}{1+3} + \frac{1^3+2^3+3^3}{1+3+5} + ...$find T_n & S_n.

Ans $T_n = \frac{n^2}{4} + \frac{n}{2} + \frac{1}{4}$; $S_n = \frac{1}{4}.\frac{n(n+1)(2n+1)}{6} + \frac{1}{2}.\frac{n(n+1)}{2} + \frac{1}{4}.n$

Sol. $t_n = \frac{1^3+2^3+3^3+....+n^3}{1+3+5+....(2n-1)} = \frac{\left\{\frac{n(n+1)}{2}\right\}^2}{\frac{n}{2}\{2+2(n-1)\}} = \frac{\frac{n^2(n+1)^2}{4}}{n^2} = \frac{(n+1)^2}{4} = \frac{n^2}{4} + \frac{n}{2} + \frac{1}{4}$

$\therefore S_n = \Sigma t_n = \frac{1}{4}\Sigma n^2 + \frac{1}{2}\Sigma n + \frac{1}{4}\Sigma 1 = \frac{1}{4}.\frac{n(n+1)(2n+1)}{6} + \frac{1}{2}.\frac{n(n+1)}{2} + \frac{1}{4}.n$

7.2 MISCELLANEOUS SEQUENCES - TYPE 2

Let u_1, u_2, u_3 ..... be a sequence, such that $u_2 - u_1, u_3 - u_2$, ... is either an A.P. or a G.P. then nth term u_n of this sequence is obtained as follows

$S = u_1 + u_2 + u_3 + ... + u_n$ .....(i)

$S = u_1 + u_2 + ... + u_{n-1} + u_n$ .....(ii)

(i) – (ii) $\Rightarrow$ $u_n = u_1 + (u_2 - u_1) + (u_3 - u_2) + ... + (u_n - u_{n-1})$

Where the series $(u_2 - u_1) + (u_3 - u_2) + ... + (u_n - u_{n-1})$ is either in A.P. or in G.P. then we can find u_n.

So sum of series $S = \sum_{r=1}^{n} u_r$

Q.1 Find the sum to n-terms $3 + 7 + 13 + 21 + ...$

Ans $\frac{n}{3}(n^2 + 3n + 5)$

Sol. Let $S = 3 + 7 + 13 + 21 + ... + T_n$ .....(i)

$S = 3 + 7 + 13 + ... + T_{n-1} + T_n$ .....(ii)

(i) – (ii) $\Rightarrow$

$T_n = 3 + 4 + 6 + 8 + ... + (T_n - T_{n-1})$

$= 3 + \frac{n-1}{2}[8 + (n-2)2]$

$= 3 + (n-1)(n+2)$

$= n^2 + n + 1$

Hence $S = \sum (n^2 + n + 1)$

$= \sum n^2 + \sum n + \sum 1$

$= \frac{n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2} + n = \frac{n}{3}(n^2 + 3n + 5)$

Q.2 $S = 6 + 13 + 22 + 33 + 46 + \ldots\ldots\ldots\ldots T_n$ Find T_n

Ans. $T_n = n^2 + 4n + 1$

Sol. $S = 6 + 13 + 22 + 33 + 46 + \ldots\ldots\ldots\ldots T_n$

$S = 6 + 13 + 22 + 33 + \ldots\ldots\ldots\ldots T_{n-1} + T_n$

$0 = 6 + 7 + 9 + 11 + 13 + \ldots\ldots\ldots (T_{n-1} - T_n) - T_n$

$T_n = 6 + 7 + 9 + 11 + 13 + \ldots\ldots\ldots\ldots (T_{n-1} - T_n)$

$T_n = 6 + \left[\frac{n-1}{2}(14 + (n-2)2)\right]$

$= 6 + [(n-1)(n+5)]$

$= 6 + n^2 + 4n - 5$

$T_n = n^2 + 4n + 1$

Q.3 $S = 2 + 6 + 12 + 20 + 30 + \ldots\ldots\ldots T_n$ Find T_n

Ans. $T_n = n^2 + n$

Sol. $S = 2 + 6 + 12 + 20 + \ldots\ldots\ldots T_{n-1} + T_n$

$T_n = 2 + 4 + 6 + 8 + 10 + \ldots\ldots\ldots (T_{n-1} - T_n) T_n$

$T_n = \frac{n}{2}(4 + (n-1)2)$

$= \frac{n}{2}(2n + 2) \Rightarrow n(n+1) \Rightarrow n^2 + n$

Q.4 Find the sum to n-terms $1 + 4 + 10 + 22 + \ldots\ldots\ldots$

Ans $3.2^n - 2n - 3$

Sol. Let $S = 1 + 4 + 10 + 22 + \ldots\ldots\ldots + T_n$ .....(i)

$S = 1 + 4 + 10 + \ldots\ldots\ldots + T_{n-1} + T_n$ .....(ii)

(i) – (ii) $\Rightarrow T_n = 1 + (3 + 6 + 12 + \ldots\ldots\ldots + T_n - T_{n-1})$

$T_n = 1 + 3\left(\frac{2^{n-1} - 1}{2 - 1}\right)$

$T_n = 3 \cdot 2^{n-1} - 2$

So $S = \sum T_n = 3 \sum 2^{n-1} - \sum 2$

$$= 3 \cdot \left(\frac{2^n - 1}{2 - 1}\right) - 2n \quad = 3.2^n - 2n - 3$$

Q.5 $S = 3 + 8 + 15 + 24 + \ldots\ldots\ldots T_n$ Find T_n

Ans. $T_n = n^2 + 2n$

Sol. $S = 3 + 8 + 15 + 24 + \ldots\ldots\ldots T_n$

$S = 3 + 8 + 15 + \ldots\ldots\ldots T_{n-1} + T_n$

$T_n = 3 + 5 + 7 + 9 + \ldots\ldots\ldots (T_{n-1} - T_n)$

$T_n = \frac{n}{2}[6 + (n-1)2] \Rightarrow n(3 + n - 1)$

$= n(n+2) \Rightarrow n^2 + 2n$

Q.6 $S = 5 + 7 + 13 + 31 + 85 + \ldots\ldots\ldots T_n$ Find T_n

Ans. $T_n = 4 + 3^{n-1}$

Sol. $S = 5 + 7 + 13 + 31 + 85 + \ldots\ldots\ldots T_n$

$S = 5 + 7 + 13 + 31 + \ldots\ldots\ldots T_{n-1} + T_n$

$T_n = 5 + 2 + 6 + 18 + 54 + \ldots\ldots\ldots (T_{n-1} - T_n)$

$T_n = 5 + \frac{2\left(3^{n-1} - 1\right)}{3 - 1}$

$T_n = 3^{n-1} + 4$

Q.7 $S = 2 + 5 + 14 + 41 + 122 \ldots\ldots\ldots T_n$ Find T_n

Ans. $T_n = \frac{3^n + 1}{2}$

Sol. $S = 2 + 5 + 14 + 41 + 122 \ldots\ldots\ldots T_n$

$S = 2 + 5 + 14 + 41 \ldots\ldots\ldots T_{n-1} + T_n$

$T_n = 2 + 3 + 9 + 27 + 81 \ldots\ldots\ldots (T_{n-1} - T_n)$

$= 2 + \frac{3\left(3^{n-1} - 1\right)}{2}$

$$= \frac{4+3^n-3}{2} \Rightarrow \frac{3^n+1}{2}$$

Method of difference for finding S_n

If possible express r^{th} term as difference of two terms as $t_r = \pm (f(r) - f(r \pm 1))$. This can be explained with the help of examples given below.

$t_1 = f(1) - f(0)$,
$t_2 = f(2) - f(1)$,
$\vdots \quad \vdots \qquad \vdots$
$t_n = f(n) - f(n-1)$
$\Rightarrow \quad S_n = f(n) - f(0)$

7.3 MISCELLANEOUS SEQUENCES - TYPE 3

If each term is compared of product of r factors in AP in denominator and first factor of each term will also be in AP.

Q.1 $\frac{1}{1.2.3} + \frac{1}{2.3.4} + \frac{1}{3.4.5} + \ldots\ldots\ldots\ldots$ find S_n and S_∞

Ans. $S_n = \frac{1}{2}\left(\frac{1}{2} - \frac{1}{(n+1)(n+2)}\right)$; $S_\infty = \frac{1}{4}$

Sol a) Write T_n

$$T_n = \frac{1}{n(n+1)(n+2)}$$

b) Now check following two points
(i) Factors in denominator are in AP
(ii) Numerator is constant.

c) Express numerator as diff. of extreme factors

$$T_n = \frac{1}{2}\left[\frac{(n+2)-n}{n(n+1)(n+2)}\right] = \frac{1}{2}\left[\frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)}\right]$$

Q.2 $S = \frac{1}{1.3.5} + \frac{1}{3.5.7} + \frac{1}{5.7.9} + \ldots\ldots\ldots$find S_n & S_∞.

Ans. $S_n = \frac{1}{4}\left(\frac{1}{1.3} - \frac{1}{(2n+1)(2n+3)}\right)$; $S_\infty = \frac{1}{12}$

Sol. $T_r = \frac{1}{(2r-1)(2r+1)(2r+3)} \Rightarrow \frac{1}{4}\left[\frac{(2r+3)-(2r-1)}{(2r-1)(2r+1)(2r+3)}\right]$

$$T_r = \frac{1}{4}\left[\frac{1}{(2r-1)(2r+1)} - \frac{1}{(2r+1)(2r+3)}\right]$$

$$T_1 = \frac{1}{4}\left[\frac{1}{1\cdot 3} - \frac{1}{3\cdot 5}\right]$$

$$T_2 = \frac{1}{4}\left[\frac{1}{3\cdot 5} - \frac{1}{8\cdot 7}\right]$$

$$T_n = \frac{1}{4}\left[\frac{1}{(2n-1)(2n+1)} - \frac{1}{(2n+1)(2n+3)}\right]$$

$$S_n = \frac{1}{4}\left[\frac{1}{3} - \frac{1}{(2n+1)(2n+3)}\right]$$

$$S_\infty = \frac{1}{12}$$

Q.3 $S = \frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \ldots\ldots\ldots$find S_n & S_∞.

Ans. $S_n = \frac{n}{n+1}$; $S_\infty = 1$

Sol. $S = \frac{1}{1\cdot 2} + \frac{1}{2\cdot 3} + \frac{1}{3\cdot 4} + \ldots\ldots \frac{1}{(n-1)(n)}$

$$S_n = \frac{1}{1} - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \frac{1}{3} - \frac{1}{4} \cdots + \frac{1}{n-1} - \frac{1}{n}$$

$$S_n = 1 - \frac{1}{n} \Rightarrow \frac{n-1}{n}$$

$$S_\infty = 1$$

Q.4 $S = \frac{1}{1.3.5} + \frac{1}{2.4.6} + \frac{1}{3.5.7} + \ldots\ldots\ldots$find S_n & S_∞.

Ans. $S_n = \frac{1}{4}\left(\frac{1}{1.3} + \frac{1}{2.4} - \frac{1}{(n+1)(n+3)} - \frac{1}{(n+2)(n+4)}\right)$; $S_\infty = \frac{11}{96}$

Sol. $T_n = \frac{1}{n(n+2)(n+4)}$

$$T_n = \frac{1}{4}\left[\frac{(n+4)-n}{n(n+2)(n+4)}\right]$$

$$T_n = \frac{1}{4}\left[\frac{1}{n(n+2)} - \frac{1}{(n+2)(n+4)}\right]$$

$$T_1 = \frac{1}{4}\left[\frac{1}{1\cdot 3} - \frac{1}{3\cdot 5}\right]$$

$$T_2 = \frac{1}{4}\left[\frac{1}{2\cdot 4} - \frac{1}{4\cdot 6}\right]$$

$$T_3 = \frac{1}{4}\left[\frac{1}{3\cdot 5} - \frac{1}{5\cdot 7}\right]$$

$$\vdots$$

$$T_n = \frac{1}{4}\left[\frac{1}{n(n+2)} - \frac{1}{(n+2)(n+4)}\right]$$

$$S_n = \frac{1}{4}\left[\frac{1}{1\cdot 3} + \frac{1}{2\cdot 4} - \frac{1}{(n+2)(n+4)} - \frac{1}{(n+1)(n+3)}\right]$$

$$S_\infty = \frac{1}{4}\left[\frac{1}{3} + \frac{1}{8}\right] \Rightarrow \frac{11}{96}$$

Q.5 $T_n = \dfrac{1}{n(n+1)(n+3)}$ find S_∞

Ans. $\dfrac{7}{36}$

Sol $= \dfrac{n+2}{n(n+1)(n+2)(n+3)}$

$$= \frac{n}{n(n+1)(n+2)(n+3)} + \frac{2}{n(n+1)(n+2)(n+3)}$$

$$T_n = \underbrace{\frac{1}{(n+1)(n+2)(n+3)}}_{S_1} + \underbrace{\frac{2}{n(n+1)(n+2)(n+3)}}_{S_2}$$

Q.6 $T_n = \dfrac{n+2}{n(n+1)(n+3)}$ find S_∞

Ans. $\dfrac{29}{36}$

Sol $= \dfrac{(n+2)^2}{n(n+1)(n+2)(n+3)}$

$$= \frac{n^2+4n+4}{n(n+1)(n+2)(n+3)}$$

$$= \frac{n(n+3)+(n+3)+1}{n(n+1)(n+2)(n+3)}$$

$$T_n = \underbrace{\frac{1}{(n+1)(n+2)}}_{S_1} + \underbrace{\frac{1}{n(n+1)(n+2)}}_{S_2} + \underbrace{\frac{1}{n(n+1)(n+2)(n+3)}}_{S_3}$$

7.4 MISCELLANEOUS SEQUENCES - TYPE 4

Each term is composed of product of r factor in AP and first factor of each term will also be in AP.

Q.1 $S = 1.2.3.4 + 2.3.4.5 + 3.4.5.6 + ...$n terms

Sol $T_n = n(n+1)(n+2)(n+3)$

$$= \frac{1}{5} n(n+1)(n+2)(n+3)\ [(n+4) \quad - \quad (n-1)]$$

term next to last term term before first term

$$= \frac{1}{5}[n(n+1)(n+2)(n+3)(n+4) - (n-1)n(n+1)(n+2)(n+3)]$$

$$T_1 = \frac{1}{5}[1\cdot2\cdot3\cdot4\cdot5 - 0]$$

$$T_2 = \frac{1}{5}[2\cdot3\cdot4\cdot5\cdot6 - 1\cdot2\cdot3\cdot4\cdot5]$$

$\vdots$

$$T_n = \frac{1}{5}[n(n+1)(n+2)(n+3)(n+4) - (n-1)n(n+1)(n+2)(n+3)]$$

$$S_n = \frac{1}{5}[n(n+1)(n+2)(n+3)(n+4)]$$

Q.2 Find the sum of n terms of the series $1 . 3 . 5 + 3 . 5 . 7 + 5 . 7 . 9 + ...$

Ans $n(2n^3 + 8n^2 + 7n - 2)$

Sol. The nth term is $(2n-1)(2n+1)(2n+3)$

$T_n = (2n-1)(2n+1)(2n+3)$

$$T_n = \frac{1}{8}(2n-1)(2n+1)(2n+3)\ \{(2n+5) - (2n-3)\}$$

$$= \frac{1}{8}(V_n - V_{n-1}) \qquad [\text{Let } V_n = (2n-1)(2n+1)(2n+3)(2n+5)]$$

$$S_n = \sum T_n = \frac{1}{8}[V_n - V_0]$$

$$\therefore S_n = \frac{(2n-1)(2n+1)(2n+3)(2n+5)}{8} + \frac{15}{8} = n(2n^3 + 8n^2 + 7n - 2)$$

7.5 MISCELLANEOUS QUESTIONS

1. $T_r = \dfrac{r}{1+r^2+r^4}$ find S_n.

Ans. $S_n = \dfrac{1}{2}\left(1-\dfrac{1}{n^2+n+1}\right)$

Sol. $T_r = \dfrac{r}{(r+1)^2-r^2}$

$$T_r = \frac{2r}{2\left(r^2+r+1\right)\left(r^2-r+1\right)}$$

$$= \frac{1}{2}\left[\frac{\left(r^2+r+1\right)-\left(r^2-r+1\right)}{\left(r^2+r+1\right)\left(r^2-r+1\right)}\right]$$

$$T_r = \frac{1}{2}\left[\frac{1}{r^2-r+1}-\frac{1}{r^2+r+1}\right]$$

$$T_1 = \frac{1}{2}\left[\frac{1}{1}-\frac{1}{3}\right]$$

$$T_2 = \frac{1}{2}\left[\frac{1}{3}-\frac{1}{7}\right]$$

$$T_3 = \frac{1}{2}\left[\frac{1}{7}-\frac{1}{13}\right]$$

$\vdots$

$$T_n = \frac{1}{2}\left[\frac{1}{n^2-n+1}-\frac{1}{n^2+n+1}\right]$$

$$S_n = \frac{1}{2}\left[1-\frac{1}{n^2+n+1}\right]$$

2. $\dfrac{1}{1.3}+\dfrac{2}{1.3.5}+\dfrac{3}{1.3.5.7}+\dfrac{4}{1.3.5.7.9}+\ldots\ldots\ldots+$

Ans. $S_n = \dfrac{1}{2}\left(1-\dfrac{1}{1.3.5\ldots\ldots\ldots(2n+1)}\right)$

Sol. $T_r = \dfrac{r}{1\cdot3\cdot5\ldots(2r+1)}$

$$T_r = \frac{1}{2}\left[\frac{2r}{1\cdot3\cdot5\ldots(2r+1)}\right]$$

$$= \frac{1}{2}\left[\frac{(2r+1)-1}{1\cdot3\cdot5\ldots(2r+1)}\right]$$

$$= \frac{1}{2}\left[\frac{2r+1}{1\cdot3\cdot5\ldots(2r+1)} - \frac{1}{1\cdot3\cdot5\ldots(2r+1)}\right]$$

$$T_r = \frac{1}{2}\left[\frac{1}{1\cdot3\cdot5\ldots(2r-1)} - \frac{1}{1\cdot3\cdot5\ldots(2r+1)}\right]$$

$$T_1 = \frac{1}{2}\left[1 - \frac{1}{1\cdot3}\right]$$

$$T_2 = \frac{1}{2}\left[\frac{1}{1\cdot3} - \frac{1}{1\cdot3\cdot5}\right]$$

$$T_3 = \frac{1}{2}\left[\frac{1}{1\cdot3\cdot5} - \frac{1}{1\cdot3\cdot5\cdot7}\right]$$

$\vdots$

$$T_n = \frac{1}{2}\left[\frac{1}{1\cdot3\cdot5\ldots(2n-1)} - \frac{1}{1\cdot3\cdot5\cdot7\ldots(2n+1)}\right]$$

$$S_n = \frac{1}{2}\left[1 - \frac{1}{1\cdot3\cdot5\cdot7\ldots(2n+1)}\right]$$

3. Find S_n and S_∞ for $\frac{1}{2\cdot4} + \frac{1\cdot3}{2\cdot4\cdot6} + \frac{1\cdot3\cdot5}{2\cdot4\cdot6\cdot8} + \ldots\ldots\ldots$

Ans $S_n = \frac{1}{2} - \frac{1}{2^n}\left(\frac{1\cdot3\cdot5\ldots\ldots(2n+1)}{1.2.3\ldots\ldots(2n+1)}\right)$; $S_\infty = \frac{1}{2}$

Sol $T_n = \frac{1\cdot3\cdot5\ldots\ldots(2n-1)[(2n+2)-(2n+1)]}{2\cdot4\cdot6\ldots\ldots\ldots(2n+2)}$

$$= \frac{1\cdot3\cdot5\ldots\ldots(2n-1)}{2\cdot4\cdot6\ldots\ldots\ldots2n} - \frac{1\cdot3\cdot5\ldots\ldots(2n+1)}{2\cdot4\cdot6\ldots\ldots\ldots(2n+2)}$$

$$= \left(\frac{1}{2} - \frac{1.3}{2.4}\right) + \left(\frac{1.3}{2.4} - \frac{1.3.5}{2.4.6}\right) + \ldots\ldots\ldots\ldots + \left(\frac{1.3\ldots\ldots(2n-1)}{2.4.6\ldots\ldots2n}\right) - \frac{1.3.5\ldots\ldots(2n+1)}{2.4.6\ldots\ldots(2n+2)}$$

$$S_n = \frac{1}{2} - \frac{1\cdot3\cdot5\ldots\ldots(2n+1)}{2\cdot4\cdot6\ldots\ldots\ldots(2n+2)},$$

$$S_n = \frac{1}{2} - \frac{1}{2^{n+1}}\left(\frac{1\cdot3\cdot5\ldots\ldots(2n+1)}{1.2.3\ldots\ldots\ldots(n+1)}\right)$$

$$S_\infty = \frac{1}{2}$$

4. Let $\{a_n\}_{n=1}^{\infty}$ be a sequence such that $a_1 = 1$, $a_2 = 1$ and $a_{n+2} = 2a_{n+1} + a_n$ for all $n \geq 1$. Then the value of $47\sum_{n=1}^{\infty}\frac{a_n}{2^{3n}}$ is equal to _____. **[JEE(Main)July2021]**

माना $\{a_n\}_{n=1}^{\infty}$ एक अनुक्रम है, जिसके लिए $a_1 = 1$, $a_2 = 1$ तथा $a_{n+2} = 2a_{n+1} + a_n$ $\forall$ $n \geq 1$ हैं। तो $47\sum_{n=1}^{\infty}\frac{a_n}{2^{3n}}$ बराबर है _____ **[JEE(Main)July2021]**

Ans. 7

Sol. $a_{n+2} = 2a_{n+1} + a_n$

$$\frac{a_{n+2}}{8^n} = 2\frac{a_{n+1}}{8^n} + \frac{a_n}{8^n}$$

$$64\sum_{n=1}^{\infty}\left(\frac{a_{n+2}}{8^{n+2}}\right) = 16\sum_{n=1}^{\infty}\left(\frac{a_{n+1}}{8^{n+1}}\right) + \sum_{n=1}^{\infty}\frac{a_n}{8^n}$$

Let $\sum_{n=1}^{\infty}\left(\frac{a_n}{8^n}\right) = p$

$$64\left(p - \frac{a_1}{8} - \frac{a_2}{8^2}\right) = 16\left(p - \frac{a_1}{8}\right) + p$$

$$p = \frac{7}{47}$$

$$47\,p = 7$$